Knowledge-Aided Space-Time Adaptive Processing for Functional Ultrasound: Robust Microvascular Detection in High-Clutter Regimes

Arthur Shune*
Independent Researcher

Abstract

Functional ultrasound (fUS) and ultrafast Doppler imaging face severe interference from tissue motion, phase aberrations, and aliased high-velocity flow, which can obscure slow microvascular dynamics. Low-rank clutter suppression methods such as motion-compensated singular value decomposition (MC–SVD) are widely used, but can degrade in extreme clutter and in the far right tail of flow evidence.

Motivated by airborne radar and RF array processing, we introduce a space–time adaptive processing (STAP) framework for fUS slow-time flow detection. The method operates on localized spatiotemporal tiles with a fixed-parameter operating profile (tile geometry, slow-time aperture, Doppler band geometry, and covariance conditioning) and produces a matched-subspace STAP score. An optional knowledge-aided (KA) spatial-spectral prior uses local band-energy features to apply a conservative, shrink-only penalty in high-risk regions while protecting a high-confidence set; by construction, the post-KA score is non-increasing on penalized pixels. We also study conditional execution as a decision-rule variant that skips STAP outside a flow-proxy region and quantify its leakage sensitivity.

Quantitative detection performance is evaluated using a strict low-FPR operating discipline (typically 10^{-4}–10^{-3}, subject to empirical support), reflecting the sensitivity of downstream functional mapping to rare flow artifacts and consistent with tail-rate control philosophies in radar detection and tracking. We report results on labeled k-Wave brain simulations and on structurally labeled *in vitro* phantom acquisitions (Gammex; labels derived from B-mode geometry), and we benchmark against MC–SVD, robust PCA, and tensor HOSVD baselines as well as conventional short-ensemble Doppler scores. We further report labeled within-ensemble motion ladders on phantom data via per-shot motion injection, and label-free stability under controlled motion corruption on kilohertz ultrasound localization microscopy (ULM) IQ blocks. Because most open in-vivo fUS releases do not preserve raw high-PRF complex IQ with independent H_1/H_0 labels, we restrict quantitative ROC claims to labeled simulations and structurally labeled phantoms, using in-vivo data only for robustness and telemetry audits. All experiments and figures are accompanied by a command-level reproducibility manifest in the Supplementary Material (exact scripts, arguments, and expected artifact paths).

*Work performed in personal capacity. Correspondence: `arthur.shune@gmail.com` (alt: `arthur@skymesasystems.com`). ORCID: `0009-0004-6433-3856`. Current employment: BlackSky, Inc. Founder: SkyMesa Systems; Valence Intelligence Systems. No external funding.

1 Introduction

Microvascular flow detection in ultrasound is an extreme-tail problem: flow evidence occupies a small fraction of pixels while tissue motion, phase aberration, and aliased high-velocity clutter induce heavy-tailed background statistics. In this regime, improvements in global summaries such as AUC can be operationally uninformative; the key requirement is stable detection at ultra-low false-positive rates.

We address this setting with a localized, STAP-inspired detector in which adaptation is temporal (slow-time), while spatial localization enters through tile-local covariance estimation. Instead of relying only on global low-rank suppression, we whiten local slow-time snapshots and score energy in a flow subspace, targeting spatially varying interference patterns that dominate extreme-tail false alarms.

We further introduce a knowledge-aided (KA) shrink-only regularization layer driven by local band-energy features. KA can suppress high-risk tail events without score inflation: post-KA scores are pointwise non-increasing, and a protected set is explicitly preserved.

Contributions.

- A tile-based temporal STAP detector with matched-subspace interpretation, evaluated at strict low-FPR operating points.
- A feature-conditioned KA regularizer with shrink-only monotonicity and protected-set invariance.
- A leakage-controlled conditional-execution variant treated explicitly as a changed decision rule (not a cost-free speedup).
- Regime-stratified evidence across labeled simulations, structurally labeled Gammex flow phantom sequences, and label-free instrumentation datasets.

Scope and claims. Claims are stratified by data regime and summarized in Table 2. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels; unlabeled in-vivo IQ and PD-only datasets are used for instrumentation and robustness analyses.

Organization. Section 2 defines the detection problem and score taxonomy. Section 3 presents the localized detector and KA regularization. Section 4 specifies fixed-profile experimental design and integrity controls. Section 5 reports results, followed by discussion and conclusion.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

STAP and knowledge-aided STAP. Classical MVDR/LCMV/STAP and knowledge-aided variants shape covariances using external structure [42–44, 49, 50, 53]. Our spatial-spectral prior formalizes when such transforms are allowed to act in fUS and keeps them inert when data are out of regime.

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6,7,11,12,16]. Robust PCA and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [22, 36]. We benchmark STAP against MC–SVD, RPCA, and HOSVD baselines in brain-like regimes and report conventional short-ensemble Doppler comparators on phantom data.

Relation to eigen-based and local adaptive clutter filtering. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13–15]. Our contribution differs in the *detector* formulation and evaluation discipline: we whiten slow-time snapshots using tile-local conditioned covariances and score energy in an explicitly defined flow subspace (matched-subspace statistic), evaluate at empirically supported ultra-low FPR operating points, and treat compute gating (conditional execution) as a distinct decision rule with leakage controls.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [35]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy KA hygiene regime where the prior penalty is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [37,38]; our hemodynamic STAP variant reuses the same banded temporal machinery on PD time series and adds an alias-aware shrink-only prior penalty as a safety layer.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to STAP.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness proxy (used only for deterministic preprocessing such as

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline right-tail detector score on X_{base}	baseline ROC / low-FPR operating-point reporting	IQ
$S_{\text{stap,pre}}$	pre-KA matched-subspace STAP score	primary STAP ROC / low-FPR operating-point reporting	IQ + local covariance
$S_{\text{stap,post}}$	post-KA shrink-only score $wS_{\text{stap,pre}}$	KA tail-hygiene analyses	IQ + feature map
$\text{PD}^{\uparrow}$, proxy masks	baseline PD-brightness proxy and derived masks	gating, protected sets, visualization	IQ (or PD-only proxy mode)

Table 1: Score taxonomy used throughout. Detector scores are separated from proxy maps used for gating/diagnostics to avoid circular reporting.

proxy masks),

$$\text{PD}^{\uparrow}(y,x) = \frac{1}{T}\sum_{t=0}^{T-1} |X_{\text{base}}(t,y,x)|^2 .$$

For quantitative detection and thresholding we operate strictly on the right tail of score distributions and denote the baseline right-tail score as $S_{\text{base}}(y,x)$ (power Doppler on the baseline-filtered cube). For short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\text{PD}^{\uparrow}$ is the same underlying power statistic as S_{base} (up to fixed scaling); we keep a separate name to enforce the policy that proxy masks are used only for deterministic preprocessing and diagnostics, not as quantitative efficacy claims.

2.3 Score taxonomy

Unless explicitly stated, "STAP" refers to the pre-KA matched-subspace score $S_{\text{stap,pre}}$; KA is analyzed as an optional, shrink-only post-processing regularizer.

3 Method

Algorithm 1 (KA–STAP pipeline, fixed profile).

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y,x)$.
2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ (spatial sampling; optional Hankelization), estimate conditioned covariance R_i, and whiten $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute the matched-subspace statistic $S_{\text{stap,pre}}(i) = \frac{1}{K}\sum_k y_{i,k}^{\text{H}} P_f\, y_{i,k}$ and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y,x)$.
4. Compute local features $\phi_i = [m_{\text{alias}}(i),\ r_g(i),\ c_f(i)]^{\top}$.
5. If KA enabled: determine regime $\mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$ and compute shrink-only weight $w(\phi_i) \in (0,1]$ (protected-set invariance); output $S_{\text{stap,post}}(y,x) = w\, S_{\text{stap,pre}}(y,x)$ and audit logs (region, reason, risk mode).
6. Else: output $S_{\text{stap,post}}(y,x) = S_{\text{stap,pre}}(y,x)$.

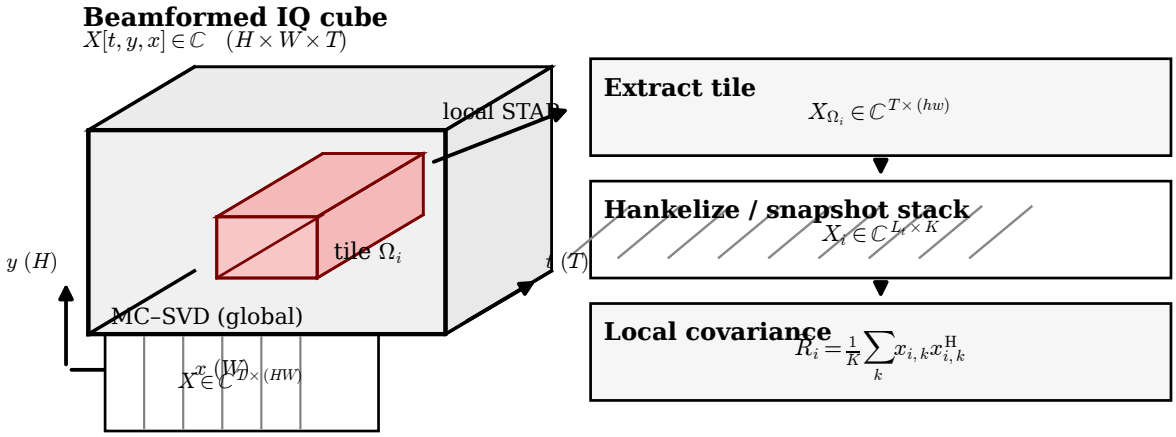

Figure 1: Schematic of spatiotemporal Hankelization and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

In practice STAP is intended as a second-stage residual covariance whitener: a coarse baseline clutter filter removes gross stationary tissue clutter, and STAP targets the remaining structured, spatially varying interference that dominates extreme-tail false alarms. A STAP-only ablation on phantom data is provided in Supplementary Material.

3.1 Temporal STAP core and Doppler band geometry

Tile-based STAP. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we apply a localized temporal STAP core on overlapping spatial tiles. We use "STAP" in a STAP-inspired sense: the adaptive filter operates in slow time, while spatial localization enters through tile-local covariance adaptation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile (via spatial sampling and, when applicable, Hankelization in slow time). Here K denotes the number of training snapshots per tile; in the fully Hankelized case $K = T - L_t + 1$ per spatial sample (or a subsampled variant when training support is capped). We estimate a local covariance

$$\widehat{R}_i := \frac{1}{K} \sum_{k=1}^{K} x_{i,k} x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2} x_{i,k}$. Figure 1 illustrates the local tile extraction and (optional) Hankelization used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

Band operators and matched-subspace STAP score. We partition the Doppler spectrum using band projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace STAP test statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\text{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_f \, y_{i,k},$$

and form a pixel-level score map $S_{\text{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-KA score is the primary STAP detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near constants (e.g. $\text{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for CFAR operation; operating points are empirically calibrated on labeled H_0 sets under the fixed-profile protocol of Section 4.1.

Spatial-spectral feature vector. In addition to the detector score, we compute per-tile band energies

$$E_f(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_f \, y_{i,k},$$

$$E_g(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_g \, y_{i,k},$$

$$E_a(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_a \, y_{i,k},$$

$$E_o(i) := \frac{1}{K} \sum_{k=1}^{K} y_{i,k}^{\text{H}} P_o \, y_{i,k},$$

and a total $E_{\text{tot}}(i) := E_f(i) + E_g(i) + E_a(i) + E_o(i) + \epsilon$ with $\epsilon > 0$. We use ratios of these energies as label-free statistics for interpretation and for the KA prior, e.g.,

$$m_{\text{alias}}(i) := \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon}, \qquad m_{\text{guard}}(i) := \log \frac{E_g(i) + \epsilon}{E_f(i) + \epsilon}, \qquad r_g(i) := \frac{E_g(i)}{E_{\text{tot}}(i)}.$$

For short ensembles, we additionally compute a non-DC peak-frequency indicator from a pooled whitened tile vector $z_i := \frac{1}{K} \sum_k y_{i,k}$ and its DFT; defining $f_{\text{peak}}(i)$ as the maximizing (non-DC) frequency, we record $\pi_f(i) := \mathbf{1}\{f_{\text{peak}}(i) \in \mathcal{B}_f\}$ and summarize peak occupancy on proxy sets by averages of π_f. Together with structural proxy coverage (below), these local descriptors form a per-tile *spatial-spectral feature vector* used to construct the KA regularizer.

3.2 Feature-conditioned score regularization

While the matched-subspace statistic S_{pre} effectively isolates flow in Gaussian clutter, highly structured artifacts can still elevate false-alarm rates in the extreme right tail. To systematically suppress these artifacts, we introduce a feature-conditioned regularization layer. While we conceptually view this penalty as a prior-odds modifier in log-score space, the construction is deterministic, using local spectral features as auxiliary statistics rather than producing a calibrated probability output.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0 : \; x \sim \mathcal{CN}(0, R_i), \qquad H_1 : \; x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, the GLRT for $\sigma_s^2 > 0$ versus $\sigma_s^2 = 0$ is monotone with the whitened subspace energy

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is a standard matched-subspace / adaptive energy detector. [51,52] Averaging $T(x)$ over the K local snapshots yields our pre-KA tile statistic $S_{\text{stap,pre}}(i)$.

Local spatial-spectral feature vector. For each tile i, we define a feature vector $\phi_i \in \mathbb{R}^3$ that summarizes local spectral kinematics and structural proxy support:

$$\phi_i := \begin{bmatrix} m_{\text{alias}}(i) \\ r_g(i) \\ c_f(i) \end{bmatrix},$$

where $m_{\text{alias}}(i) = \log \frac{E_a(i)+\epsilon}{E_f(i)+\epsilon}$ is the alias-to-flow energy ratio, $r_g(i) = E_g(i)/E_{\text{tot}}(i)$ is the guard fraction, and $c_f(i)$ is a deterministic flow-proxy coverage fraction defined below.

Feature-conditioned shrinkage (log-score view). We implement knowledge aid via a bounded weight $w(\phi_i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(\phi_i) \, S_{\text{stap,pre}}(i), \qquad w(\phi_i) \in (0, 1].$$

Equivalently, define a scale factor $\gamma(\phi_i) := 1/w(\phi_i) \geq 1$ and write

$$S_{\text{stap,post}}(i) = \frac{S_{\text{stap,pre}}(i)}{\gamma(\phi_i)}.$$

For brevity, we refer to $S_{\text{stap,pre}}$ and $S_{\text{stap,post}}$ as S_{pre} and S_{post} when discussing the feature-conditioned penalty and in figure captions. Equivalently, for $S_{\text{pre}}(i) > 0$ define $T_{\text{pre}}(i) := \log S_{\text{pre}}(i)$ and $T_{\text{post}}(i) := \log S_{\text{post}}(i)$, giving

$$T_{\text{post}}(i) = T_{\text{pre}}(i) + \log w(\phi_i).$$

We use this additive log-score form as an *interpretive lens*: it mirrors adding a feature-dependent penalty term to a log-likelihood score, but the construction is deterministic and uses ϕ_i as auxiliary statistics rather than producing calibrated probabilities. We constrain it to be shrink-only by enforcing $w(\phi_i) \leq 1$. For numerical stability in plots we display $\log_{10}(S + \epsilon)$ with $\epsilon > 0$.

Feature-space regimes. For auditability and conservative behavior, we partition the feature space into three regimes that control the strength of the prior penalty:

- **Uninformative regime $\mathcal{R}_0$.** If sample support is insufficient, band discretization is invalid, or risk features are effectively flat, the prior defaults to uninformative: $w(\phi_i) \equiv 1$, so $S_{\text{stap,post}} = S_{\text{stap,pre}}$.
- **Guarded regularization regime $\mathcal{R}_1$.** If features indicate uncertainty or leakage (e.g. elevated guard fractions or moderate alias ratios) but do not satisfy stronger actionability evidence, we apply a bounded penalty $w(\phi_i) \in [w_{\min}, 1]$ while preserving a protected high-confidence set.
- **Actionable alias-dominance regime $\mathcal{R}_2$ (research-only).** If features indicate alias-dominant structure and are *actionable* in the sense that they explain high-score background-tail events under proxy sets, we permit a stronger but still bounded penalty, while maintaining shrink-only and protected-set invariance.

Proxy sets (label-free) and protected-set invariance. When ground-truth H_1/H_0 labels are unavailable, we define proxy sets deterministically from a structural proxy (e.g. $\text{PD}^{\uparrow}$ or simulator geometry). When using $\text{PD}^{\uparrow}$, we define a high-quantile flow proxy mask

$$M_{\text{flow}}(y, x) := \mathbf{1}\{\text{PD}^{\uparrow}(y, x) \geq Q_{q_{\text{flow}}}(\text{PD}^{\uparrow})\},$$

a per-tile flow coverage

$$c_f(i) := \frac{1}{|\Omega_i|} \sum_{(y,x)\in\Omega_i} M_{\text{flow}}(y, x),$$

and background/flow proxy sets

$$\mathcal{T}_{\text{bg}} := \{i : \ c_f(i) \leq c_{\text{bg}}\}, \qquad \mathcal{T}_{\text{flow}} := \{i : \ c_f(i) \geq c_{\text{flow}}\}.$$

To enforce protected-set invariance, we define a protected set $\mathcal{T}_{\text{prot}}$ that includes high-confidence flow-proxy tiles (and optionally extreme-score tiles) and restrict penalization to a fixed candidate set of non-protected tiles in an uncertain score window; outside this candidate set we set $w(\phi_i) \equiv 1$. Region membership ($\phi_i \in \mathcal{R}_0/\mathcal{R}_1/\mathcal{R}_2$) is determined by deterministic integrity checks and by label-free actionability evidence that risk features are associated with high-score background tails (Supplementary Material).

Penalty mapping (summary). Let $S_{\text{pre}}(i)$ denote the (right-tail) pre-KA tile score and let $m_{\text{risk}}(i)$ denote the chosen risk feature (typically $m_{\text{alias}}(i)$; a guard-risk lane may substitute m_{guard} when guard dominates). Using frozen design quantiles $q_{\text{lo}}, q_{\text{hi}}, q_{\text{risk}}$, we define a candidate set

$$\mathcal{T}_{\text{cand}} := \Big\{i \notin \mathcal{T}_{\text{prot}} : \ Q_{q_{\text{lo}}}(S_{\text{pre}}) \leq S_{\text{pre}}(i) \leq Q_{q_{\text{hi}}}(S_{\text{pre}})\Big\},$$

set a background-proxy risk threshold $\tau_{\text{risk}} := Q_{q_{\text{risk}}}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}})$, and gate tiles by

$$g(i) = \mathbf{1}\{i \in \mathcal{T}_{\text{cand}}, \ m_{\text{risk}}(i) \geq \tau_{\text{risk}}\}.$$

On gated tiles we apply a bounded shrink factor

$$w(i) = \max(w_{\min}, \exp(-k\, u(i))), \quad u(i) = \max\left(0, \frac{m_{\text{risk}}(i) - \tau_{\text{risk}}}{\text{MAD}(m_{\text{risk}} \mid \mathcal{T}_{\text{bg}}) + \epsilon}\right),$$

and set $w(i) = 1$ for $g(i) = 0$ and for all $i \in \mathcal{T}_{\text{prot}}$. This yields $\gamma(i) = 1/w(i) \geq 1$ and $S_{\text{post}}(i) = S_{\text{pre}}(i)/\gamma(i)$.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0, 1]$ and define $S_{\text{post}}(i) = w(i)\, S_{\text{pre}}(i)$. For any threshold τ,*

$$\{S_{\text{post}} \geq \tau\} \subseteq \{S_{\text{pre}} \geq \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{post}}(i) \geq \tau$, then $S_{\text{pre}}(i) = S_{\text{post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected post-KA to satisfy $\mathbb{P}\{S_{\text{post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, STAP can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks).
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized flow and alias bands are empty, DC-only, or overlapping (violating orthogonality in the DFT-bin construction), we force $\phi_i \in \mathcal{R}_0$ (uninformative prior; $w \equiv 1$) with an explicit reason.
- **Sample-support and tail-sizing constraints.** Tail-association metrics are defined on a background-tail subset and become unstable when sample support is insufficient. We use a size-aware background-tail fraction $q^{\star}_{\text{tail}} = \max(q_{\text{tail}}, n_{\text{tail,min}}/n_{\text{bg}})$ and refuse to enter $\mathcal{R}_2$ (actionable prior) when tail sizes or proxy supports fall below deterministic floors.
- **Proxy-mask circularity control.** Proxy masks are used only for label-free band-energy aggregation, candidate/protected-set definitions, and conditional execution. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log the selected feature-space region and explicit reason labels so that "KA off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Supplementary Material provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the KA prior should be active or inert.

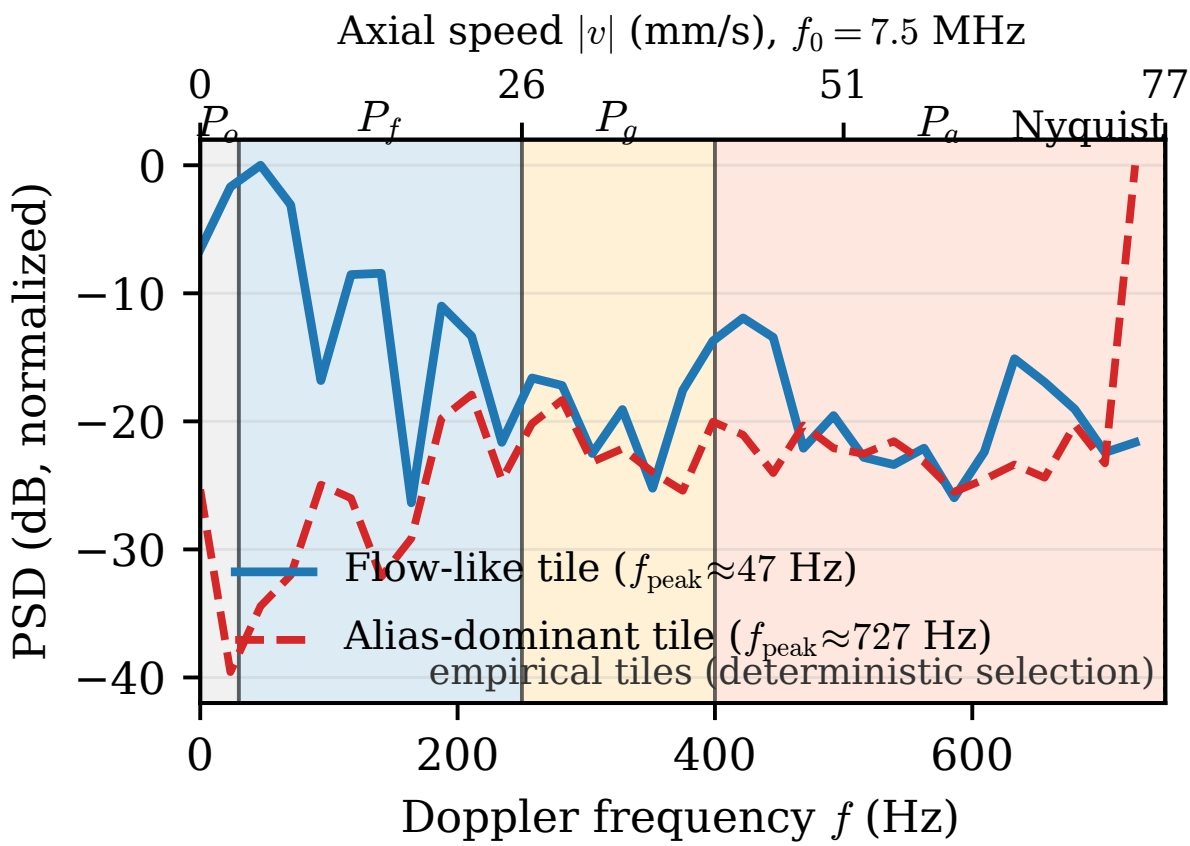

Figure 2: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on a Brain-AliasContract k-Wave pilot window (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a); the feature vector ϕ_i aggregates band energies in these regimes to drive the spatial-spectral prior.

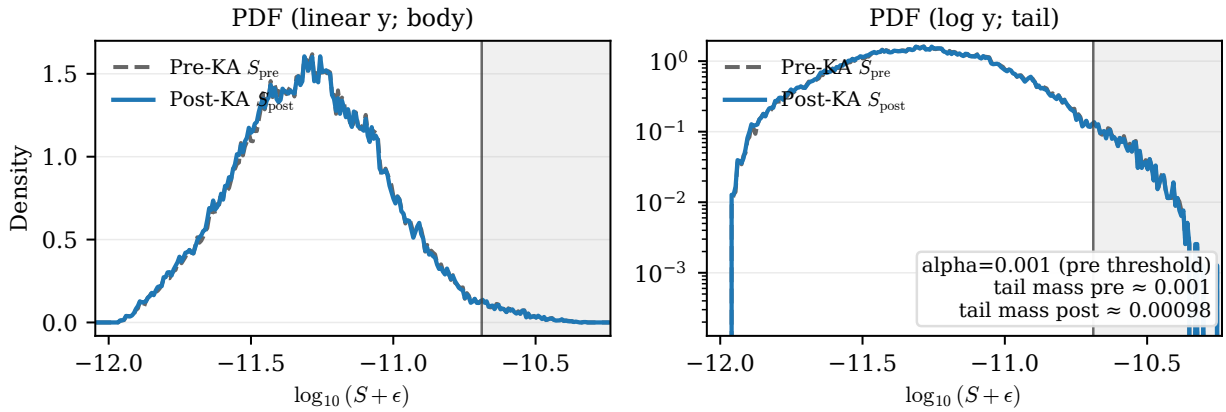

Figure 3: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution (Twinkling calculi artifact sequence). The uninformed matched-subspace statistic S_{pre} exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing penalty $\gamma(\phi_i) \geq 1$ yields the post-regularization score $S_{\text{post}} = S_{\text{pre}}/\gamma(\phi_i)$, compressing the background tail while preserving a protected high-confidence set (shrink-only and protected-set invariance). The vertical line indicates the pre-KA threshold τ_α chosen so that $\mathbb{P}\{S_{\text{pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-KA tail mass is evaluated at the same τ_α.

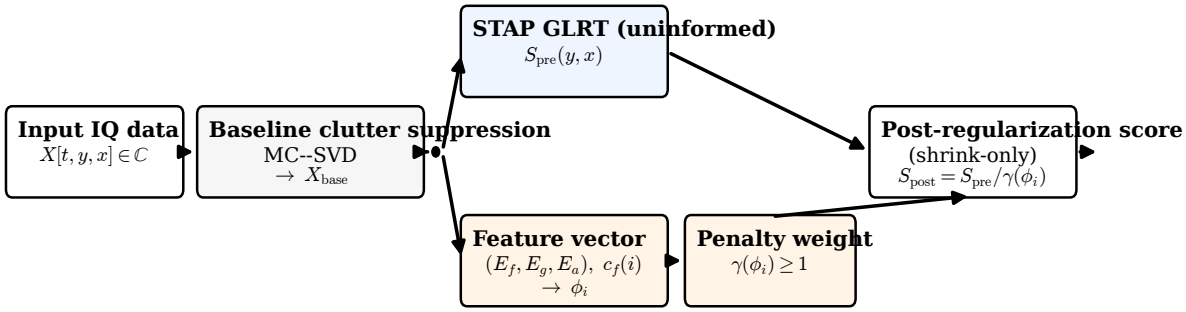

Figure 4: System block diagram for the proposed Knowledge-Aided Space-Time Adaptive Processing (KA–STAP) architecture. Beamformed kilohertz data initially undergoes baseline motion-compensated clutter filtering (MC–SVD), yielding a residual cube X_{base}. The residual is processed via two parallel branches: (top) the uninformed STAP core computes the baseline matched-subspace statistic $S_{\text{pre}}(y,x)$; (bottom) the spatial-spectral feature extractor evaluates localized band energies to construct the feature vector ϕ_i, which governs a bounded penalty $\gamma(\phi_i) \geq 1$. The branches recombine to form the strictly bounded post-regularization score $S_{\text{post}}(y,x) = S_{\text{pre}}(y,x)/\gamma(\phi_i)$ (shrink-only).

4 Experimental Design

4.1 Fixed-profile protocol and operating-point reporting

All hyperparameters (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed per regime before reporting results. Detection is evaluated in the right tail at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points.

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode → baseline → STAP score → spatial-spectral features → optional KA regularization). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the KA layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) force $\phi_i \in \mathcal{R}_0$ (uninformative prior; $w \equiv 1$) with an explicit reason label. A concise checklist (and where each sentinel is logged) is provided in Supplementary Material.

4.3 Regimes and claim scope

Claims are stratified by regime to separate quantitative detection efficacy from label-free instrumentation.

- **Brain-k-Wave simulations (labeled).** Across Brain-OpenSkull / Brain-AliasContract / Brain-SkullOR regimes, the fixed profile yields a nontrivial right-tail shoulder in the primary detector scores (baseline power Doppler and matched-subspace STAP; Notation), with the strongest low-FPR gains in the alias-augmented Brain-AliasContract and skull/OR-inspired Brain-SkullOR regimes (Table 4).
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** We decode raw beamformer outputs, derive B-mode-only tube masks (flow/background), benchmark conventional short-ensemble Doppler baselines, and report structural-label low-FPR operating points and within-ensemble motion ladders.
- **ULM Zenodo 7883227 (real kHz IQ).** We report label-free motion-robustness curves on beamformed IQ blocks using controlled rigid/nonrigid motion injection, with fairness alignment and degeneracy checks (no task ROC).
- **Supplementary analyses.** Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

5 Results

Results roadmap. We first report labeled Brain-* simulation results at ultra-low FPR, then structurally labeled Gammex flow phantom results (including motion ladders), and finally label-

Regime	Data type	Claim scope
Brain-* (k-Wave)	simulated IQ	low-FPR ROC and latency (simulation)
Gammex flow phantom	real Doppler IQ (phantoms)	structural-label ROC + motion ladders (phantom; not clinical)
ULM Zenodo 7883227	real kHz IQ	motion robustness proxies (no task ROC)
Supplementary datasets	real IQ and PD-only	descriptive/audit analyses (no task ROC)

Table 2: Claims discipline by regime. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom data (Gammex flow phantom). ULM 7883227 is used for label-free motion-robustness evidence on real kHz IQ. Additional datasets are used only for descriptive/audit analyses and are reported in Supplementary Material.

Regime (sampling)	Baseline	STAP core (fixed profile)	Doppler bands (Hz)
Brain-* (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ frozen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 3: Frozen operating profiles for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Supplementary Material.

free motion-robustness evidence on real kHz IQ (ULM 7883227). Additional dataset audits and diagnostic panels are provided in Supplementary Material.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry) using the strict low-FPR operating protocol of Section 4.1 at ultra-low FPR. Real kHz IQ datasets are used for instrumentation (band-energy integrity checks and label-free motion-robustness proxies), not task ROC. Conditional execution is a decision-rule variant (compute gating) that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.1 Brain-k-Wave regimes (labeled): low-FPR ROC and latency

We report labeled low-FPR operating points on Brain-* k-Wave simulations using right-tail detector scores (Notation): a baseline power Doppler score and the proposed matched-subspace STAP score. Baseline methods differ only in clutter suppression (MC–SVD, RPCA, HOSVD) and are scored using the same power Doppler statistic; STAP is evaluated on top of the MC–SVD residual. Across Brain-* regimes, these global baselines degrade sharply in the ultra-low-FPR tail, while the localized STAP score retains a distinct, non-saturated shoulder, including in the SkullOR regime where alias

Method / score	Brain-OpenSkull TPR@10^{-4}	Brain-OpenSkull TPR@3×10^{-4}	Brain-OpenSkull TPR@10^{-3}	Brain-AliasContract TPR@10^{-4}	Brain-AliasContract TPR@3×10^{-4}	Brain-AliasContract TPR@10^{-3}	Brain-SkullOR TPR@10^{-4}	Brain-SkullOR TPR@3×10^{-4}	Brain-SkullOR TPR@10^{-3}
MC–SVD (power Doppler)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
RPCA (power Doppler)	0.0015 (0.0000,0.0046)	0.0031 (0.0000,0.0062)	0.0062 (0.0000,0.0231)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
HOSVD (power Doppler)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0015)	0.0062 (0.0031,0.0073)	0.0015 (0.0000,0.0015)	0.0015 (0.0015,0.0046)	0.0077 (0.0069,0.0092)	0.0015 (0.0000,0.0015)	0.0015 (0.0000,0.0015)	0.0031 (0.0015,0.0031)
STAP (pre-KA matched-subspace) on MC–SVD residual	0.0223 (0.0031,0.3922)	0.2107 (0.0077,0.4610)	0.2834 (0.1496,0.5302)	0.3499 (0.2649,0.3756)	0.4033 (0.3283,0.4560)	0.5063 (0.4383,0.5113)	0.3791 (0.2622,0.3987)	0.4460 (0.3230,0.4648)	0.5102 (0.4364,0.5456)

Table 4: Brain-* k-Wave low-FPR operating points using right-tail detector scores: baseline power Doppler and matched-subspace STAP. Each regime uses five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ ($fpr_{\min} \approx 1.99 \times 10^{-5}$). Entries report median (IQR) TPR across windows at each target FPR. Baseline methods differ only in clutter suppression (MC–SVD, RPCA, HOSVD); STAP is applied on top of MC–SVD. Full per-window outputs are provided in Supplementary Material.

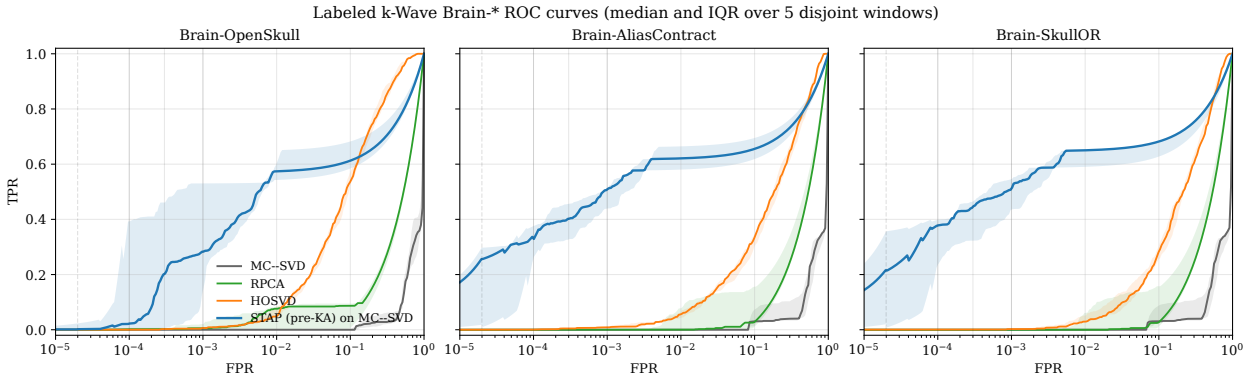

Figure 5: Brain-* k-Wave labeled regimes: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the Brain-* low-FPR operating-point reporting (Table 4). Curves show the median and interquartile range over five disjoint windows. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

deliberately leaks into the positive class. Table 4 summarizes low-FPR operating points for representative pilot realizations over five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ background pixels ($fpr_{\min} = 1/n_{\text{neg}} \approx 1.99 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically supported. Numbers are medians (IQR) over windows and are computed by choosing τ per method per window on that window's negative set to match the target FPR, then evaluating TPR on the corresponding positive set. The wider IQR at $\alpha = 10^{-4}$ in Brain-OpenSkull reflects window-to-window nonstationarity and finite tail support under a fixed, local-covariance profile (shoulder windows versus near-floor windows). Per-window thresholds and realized FPRs are provided in Supplementary Material. A cross-window threshold-transfer audit (calibrate τ_α on one window and evaluate on a disjoint window) is reported in Supplementary Material.

Baseline fairness and frozen settings. All baseline hyperparameters are fixed prior to reporting results. For Brain-* pilots, the MC–SVD energy-fraction cutoff is selected by a one-time sweep and then frozen at $e = 0.90$ for all reported windows (sweep report in `reports/brain_mcsvd_energy_sweep_seed1.csv` / `reports/brain_mcsvd_energy_sweep_seed1.json`). RPCA and HOSVD baselines use standard, reproducible settings documented in Supplementary Material. All baselines are scored using the same right-tail power Doppler statistic so that differences isolate the clutter-suppression stage rather than the scoring rule.

Latency and conditional execution. We also report representative runtimes for baseline clutter suppression and STAP under a conservative fixed operating profile (tile geometry, slow-time aperture L_t, and covariance-conditioning targets fixed a priori). Conditional execution reduces wall-clock time by skipping tiles with zero overlap with a proxy flow mask and falling back to the baseline there, but it is a *different* decision rule and can change operating characteristics; we therefore treat it as a separate algorithm variant and require explicit leakage controls when reporting conditional-execution performance.

Regime	MC–SVD (s)	STAP full (s)	STAP conditional (s)
Brain-OpenSkull	≈ 1.2	≈ 61	≈ 4.7
Brain-AliasContract	≈ 1.0	≈ 45	≈ 3.9
Brain-SkullOR	≈ 1.2	≈ 56	≈ 3.2

Table 5: Representative latency summary for the conservative reference profile on Brain-* k-Wave regimes (per 64-frame window). Conditional execution is a compute-gating variant that uses a proxy mask and changes the decision rule; it is reported as a deployment optimization requiring leakage ablations rather than as a cost-free speedup. All runs are CPU-only (`stap_device=cpu`); times are approximate wall-clock durations from runtime telemetry.

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs; we therefore treat conditional execution as a distinct decision rule rather than a cost-free speedup. Table 6 summarizes the effect on Brain-OpenSkull pilots; full bootstrap CIs and details are provided in Supplementary Material.

5.2 Phantom acquisitions (Gammex flow phantom): structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [35], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Because Doppler PRF is not reliably available in the provided metadata for these acquisitions, all Twinkling experiments treat PRF as an explicit experimental assumption frozen per sequence family (listed in Supplementary Material) and enforce band-discretization sanity checks that force $\phi_i \in \mathcal{R}_0$ when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Supplementary Material (PRF overrides and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler/STAP outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that

Conditional gating mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	+0.31	+0.81	+3.05
Disjoint-window proxy mask	−3.05	−2.90	−1.74
Random mask (matched coverage)	−3.05	−2.97	−2.36

Table 6: Conditional-execution leakage ablation on Brain-OpenSkull pilots (conditional minus full STAP; five independent realizations). Entries report mean ΔTPR at fixed FPR targets; full bootstrap CIs and details are provided in Supplementary Material.

excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 6 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences (PRF = 2500 Hz, $N = 17$ shots), we freeze a short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, STAP aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are frozen to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label low-FPR ROC (Gammex). We pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 9 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 7 summarizes the resulting structural-label operating points for the two Gammex views under their frozen profiles; full pooled summaries (including baseline-score variants and realized thresholds) are provided in Supplementary Material. Because STAP is evaluated on the same baseline-filtered residual cube, the strong STAP TPR at strict FPR rules out the interpretation that the baseline filter simply removed lumen flow; rather, the failure mode is heavy-tailed background scores dominating the right-tail threshold (see Figure 9).

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot rigid translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and STAP, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and

	Along (linear17 @ PRF 2500)			Across (linear17 @ PRF 2500)		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
Baseline (power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (log power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 power)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
STAP (pre-KA matched-subspace)	0.634 [0.628,0.640]	0.803 [0.798,0.808]	0.946 [0.943,0.948]	0.846 [0.842,0.850]	0.891 [0.888,0.893]	0.938 [0.936,0.940]

Table 7: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6\times10^{-7}$) and $n_{\mathrm{bg}} = 1.97\times10^6$ for the across view ($fpr_{\min} \approx 5.1\times10^{-7}$). Reproducibility details are provided in Supplementary Material.

fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 10 shows the resulting curves; reproducibility details are provided in Supplementary Material.

Supplementary KA hygiene (artifact-heavy calculi). Additional KA hygiene analyses on artifact-heavy calculi sequences (region/reason summaries and representative pre/post tail examples) are provided in Supplementary Material.

5.3 ULM Zenodo 7883227 (real kHz IQ): baseline calibration and nonrigid motion robustness

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [33, 34] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running baseline+STAP under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles 8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and

micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace STAP score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 11 shows the main `brainlike` sweep; additional motion variants are provided in Supplementary Material.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and STAP in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the STAP score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Supplementary Material). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

Supplementary datasets and diagnostics. Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels (e.g. conditional-execution leakage ablations, KA hygiene panels) are provided in Supplementary Material.

6 Discussion

A) Robustness in the ultra-low-FPR tail

This work adapts space–time adaptive processing (STAP), a foundational approach in radar target detection under heavy clutter, to functional ultrasound (fUS) slow-time flow detection. Under the strict low-FPR operating protocol (Section 4.1), we find that common global or semi-global clutter suppressors (MC–SVD, RPCA, HOSVD) can lose sensitivity in the extreme right tail, while the proposed tile-based matched-subspace STAP score preserves a distinct, non-saturated shoulder at ultra-low FPR. This behavior is demonstrated on labeled k-Wave simulations and structurally labeled *in vitro* phantoms, where flow/background masks are defined independently from Doppler/STAP outputs via simulator truth or B-mode geometry. The within-ensemble motion ladders on phantom acquisitions further indicate that STAP retains detection stability under controlled per-shot motion corruption, consistent with its reliance on localized covariance rather than global low-rank structure.

B) Knowledge-aided prior as a fail-safe

The optional knowledge-aided (KA) layer is designed as a conservative safety regularizer rather than an unconstrained signal amplifier. It acts as a shrink-only prior penalty with an auditable

(region, reason) output: when spatial-spectral evidence is flat or non-actionable, the prior defaults to $\mathcal{R}_0$ (uninformative; $w \equiv 1$) and leaves the pre-KA detector score unchanged. When the prior lies in $\mathcal{R}_1/\mathcal{R}_2$, it enforces protected-set invariance and applies only non-increasing shrink on high-risk regions, so that "KA does nothing" is an intended success mode out of regime. This posture prioritizes transparent regime diagnostics and avoids score inflation in regimes where spectral evidence cannot justify aggressive intervention.

C) Data limitations and the phase-history barrier

A key limitation in evaluating advanced Doppler-side covariance methods is field-wide data availability: many open-access *in vivo* fUS datasets are temporally downsampled, post-filtered, or released only as power-Doppler equivalents, which can irreversibly destroy the high-rate complex phase history needed for spatiotemporal covariance estimation and band-energy integrity checks. Accordingly, while we report real-data instrumentation (band-energy integrity, prior-regime behavior, and label-free stability under controlled motion corruption), comprehensive *in vivo* ROC evaluation of Doppler-side STAP/KA requires high-PRF raw IQ acquisitions prior to proprietary clutter suppression, with reliable metadata and task/landmark structure to define independent evaluation labels.

7 Conclusion and future work

This study establishes a deterministic, auditable KA-STAP framework for fUS slow-time flow detection under a tune-once-then-freeze discipline. Using a strict low-FPR operating protocol and independent structural labels where available, we show that localized matched-subspace STAP can preserve separability in regimes where conventional low-rank baselines degrade in the extreme tail, while the KA spatial-spectral prior provides shrink-only, protected-set-preserving risk control with explicit out-of-regime behavior.

Future work is to (i) evaluate Doppler-side STAP/KA on pre-registered, labeled raw-IQ *in vivo* datasets that preserve phase history and support independent task/landmark labels, (ii) reduce latency via hardware-aware implementations of the batched covariance and band-energy kernels toward real-time deployment, and (iii) explore compact, frozen learned regularizers only when they can be constrained to preserve the shrink-only auditability and low-FPR reporting guarantees established here.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. A complete reproduction manifest (commands, configurations, and artifact naming) is provided in Supplementary Material.

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[14] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[15] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[16] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[17] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[18] Urban, A., et al. Mobile human brain imaging using functional ultrasound. *Nature Biomedical Engineering* 8, 534–546 (2024).

[19] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[20] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[21] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[22] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[23] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[24] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[25] Functional ultrasound imaging of the brain. UNC/NCSU Joint Department of Biomedical Engineering teaching material, online resource, accessed 2024. Available at `https://bme.unc.edu/wp-content/uploads/sites/917/2021/08/functional_ultrasound.pdf`.

[26] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). Available at `https://www.frontiersin.org/articles/10.3389/fnins.2019.01384`.

[27] Rapid full-wave phase aberration correction method for transcranial ultrasound. Online resource, accessed 2024. Available at `https://jtultrasound.biomedcentral.com/articles/10.1186/s40349-016-0074-7`.

[28] Ultrafast Ultrasound Imaging. Online resource, accessed 2024. Available at `https://mdpi-res.com/bookfiles/book/756/Ultrafast_Ultrasound_Imaging.pdf`.

[29] Staggered multiple-PRF ultrafast color Doppler. Online resource, accessed 2024. Available at `https://www.biomecardio.com/publis/ieeetmi16.pdf`.

[30] Functional ultrasound imaging: a useful tool for neuroscience. Online resource, accessed 2024. Available at https://www.sciencedirect.com/science/article/pii/S1053811921009940.

[31] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[32] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* (2024). Available at https://www.nature.com/articles/s41467-024-47154-2.

[33] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[34] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: https://doi.org/10.1038/s41551-021-00824-8.

[35] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[36] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[37] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[38] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[39] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: https://doi.org/10.1002/j.1538-7305.1978.tb02104.x.

[40] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: https://doi.org/10.1109/PROC.1982.12433.

[41] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[42] Guerci, J. R. *Space-Time Adaptive Processing for Radar*. Artech House (2003).

[43] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. Space-time adaptive processing: a knowledge-based perspective for airborne radar. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: `https://doi.org/10.1109/MSP.2006.1593337`.

[44] Melvin, W. L., and Guerci, J. R. Knowledge-aided signal processing: a new paradigm for radar and other advanced sensors. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248215`.

[45] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: `https://doi.org/10.1109/53.665`.

[46] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[47] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: `https://doi.org/10.1109/36.957286`.

[48] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[49] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[50] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[51] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[52] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[53] Ward, J. Space-time adaptive processing for airborne radar. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).

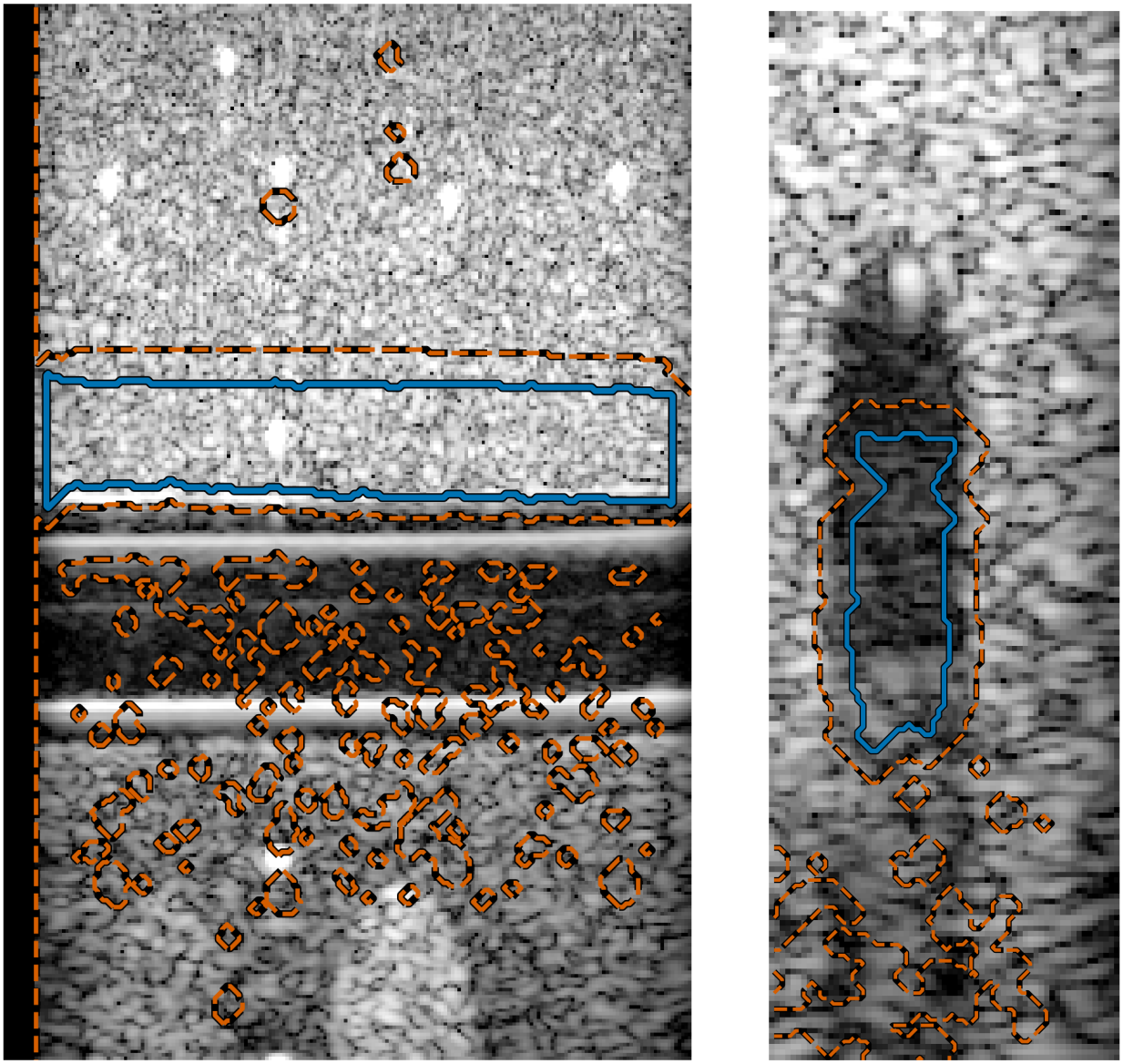

Figure 6: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

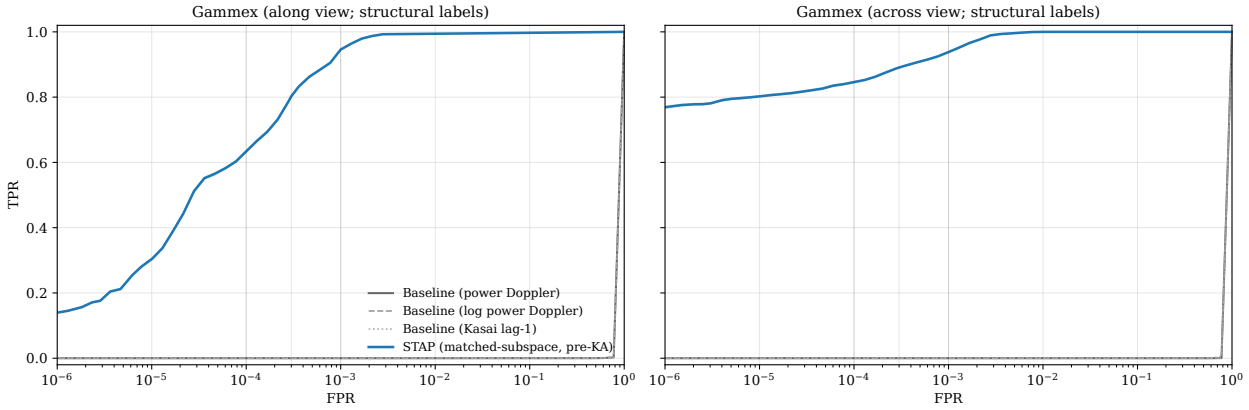

Figure 7: Gammex flow phantom structural labels: pooled ROC-style curves on along and across views, shown for baseline comparators and the pre-KA STAP matched-subspace score. The vertical guides indicate the strict low-FPR operating points reported in Table 7.

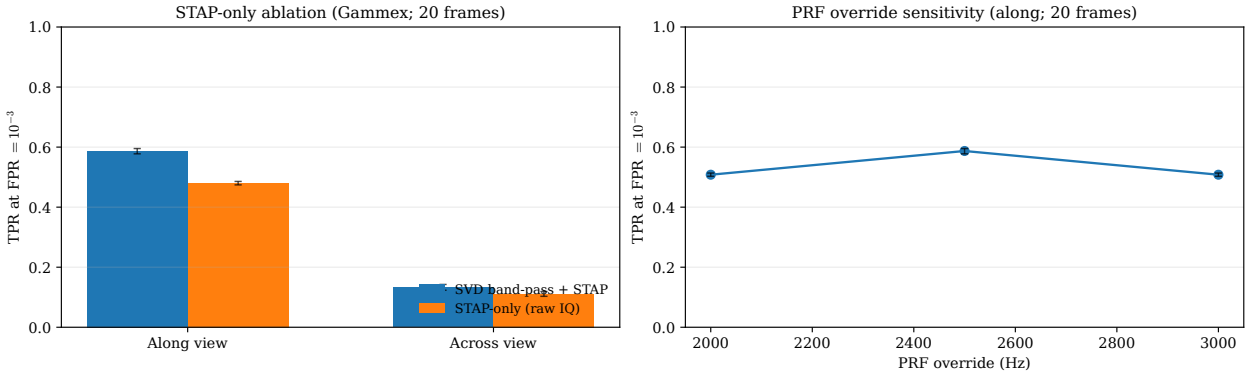

Figure 8: Gammex audits: STAP-only ablation and PRF override sensitivity (first 20 frames; along and across views). Left: STAP-only (raw IQ) vs baseline SVD band-pass + STAP at fixed FPR (10^{-3}). Right: PRF override sensitivity on the along view (PRF $\in \{2000, 2500, 3000\}$ Hz) under a frozen operating profile; this is a low-cost sanity audit of Doppler-band discretization, since PRF is constant within a sequence family and the discretization sentinels depend only on (N, PRF). Additional details are provided in Supplementary Material.

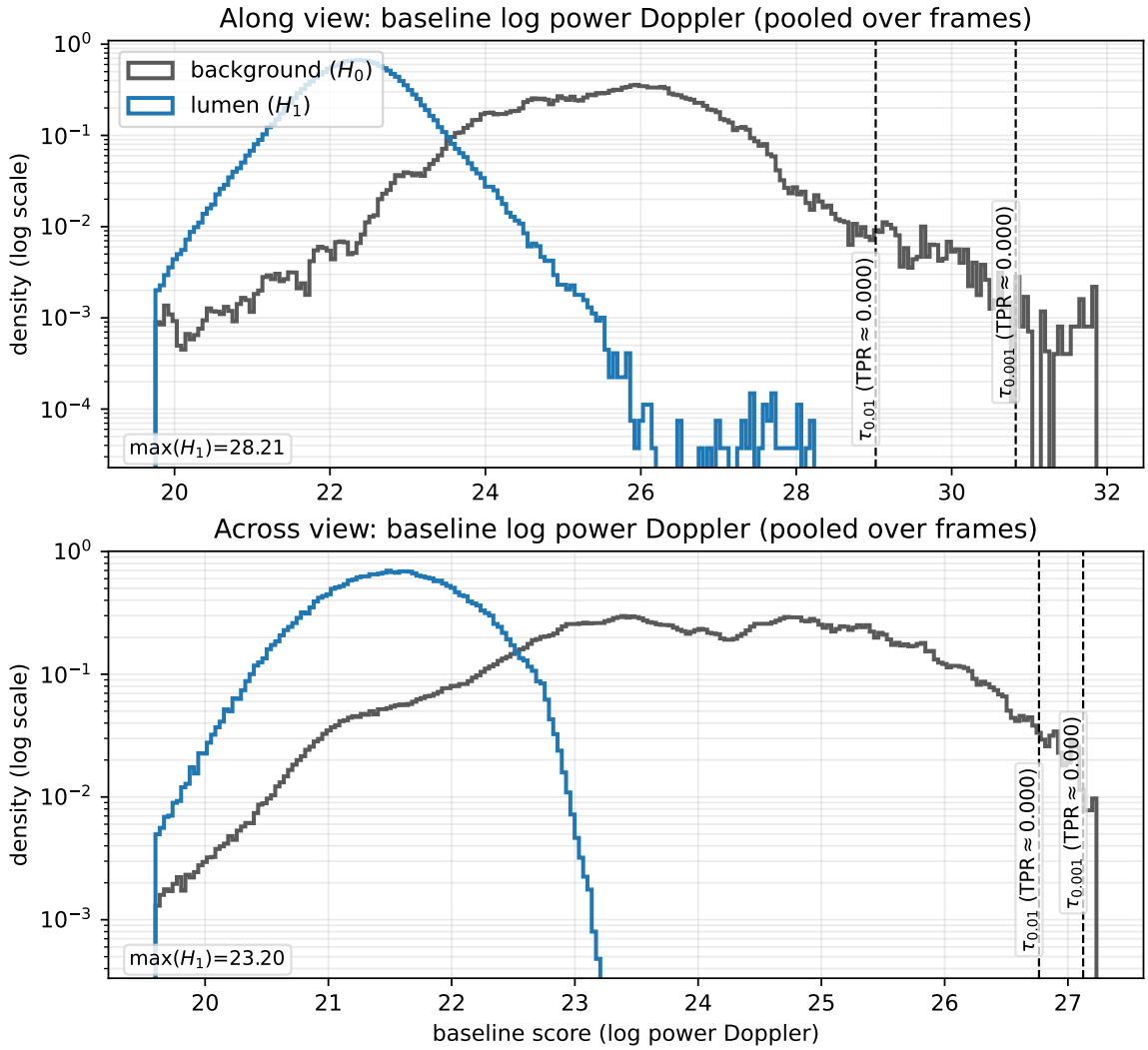

Figure 9: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

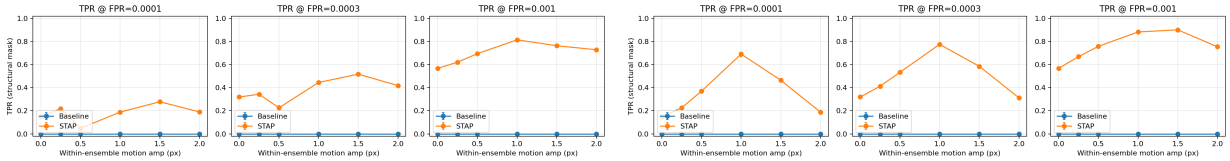

Figure 10: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each panel reports TPR vs motion amplitude at fixed FPR targets using B-mode structural masks. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Supplementary Material.

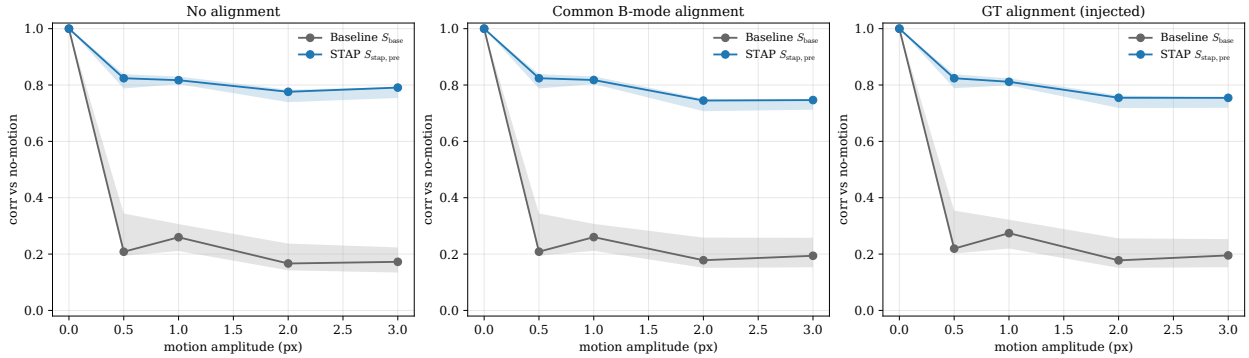

Figure 11: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the STAP score $S_{\text{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Supplementary Material.